Arijit De

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PROFESSIONAL SUMMARY

Senior AI Engineer (PhD) with 6+ years building and deploying production-grade Agentic AI systems, including multi-agent Generative AI (GenAI) platforms. Expert in Large Language Models (LLMs), Retrieval Augmented Generation (RAG), prompt engineering, and multi-agent orchestration (LangGraph, Google ADK, CrewAI, MCP) over vector databases and semantic search, using Google Vertex AI and AWS Bedrock. Builds cloud-native Python/FastAPI microservices on GCP (Cloud Run, Workflows) with LLMOps evaluation, observability, and responsible AI guardrails. Owns architecture trade-offs, effort estimation, code review, mentoring, and stakeholder POC presentations.

WORK EXPERIENCE

Senior AI Engineer | mVizn, Singapore | Remote Full time **AUG 2026 – PRESENT**

- Upgraded a computer vision system with multi-agent AI integration (**LangChain**, **LangGraph, Python, FastAPI**) for Smart Project Management, focusing on intelligent routing, smart reporting and agentic workflow execution increasing business efficiency by 70%.
- Used **Transformer** based model **finetuning** using **BERT** for intent classification and workflow routing from system signals and user queries, cost optimizing the system and reducing token usage by 35%.
- Deployed Local LLMs (**Ollama**) for secured workflow execution and data privacy in critical port systems.
- Hardened multi-agent workflows for production reliability with tool-validation layer (**Pydantic** schema + business-rule checks), agent guardrails, and human-in-the-loop approval gates for high-stakes actions cutting task failure rate by 60%.
- Translated business requirements into functional requirements along with system design for building fast, secure and cost optimized solutions.

Senior AI Engineer | Common Lab Ventures, Singapore | Remote Full time **SEP 2025 – JUL 2026**

- Developed complex multi-agent solutions with multimodal **RAG** pipelines over massive text, image, and CAD datasets (10M+ vectors in **Pinecone**) using hybrid search and cross encoder reranking, achieving retrieval accuracy (recall@k) of 92% while holding retrieval latency (TTFT) under 600 ms.
- Reduced document ingestion latency by 60% and enabled horizontal scaling for high-volume data processing by designing a distributed, event-driven **ETL** pipeline using **PySpark** and **Databricks** to process PDFs, DOCX, spreadsheets, images, and CAD files for RAG-based multi-agent workflows.
- Led the end-to-end deployment of AI services on **GCP**, establishing rigorous system evaluation benchmarks to monitor performance across 3,000+ daily user queries, ensuring high availability and 99% uptime.
- Instrumented observability via **LangSmith** tracing and **Ragas** evaluation, lifting evaluation scores by 40%.
- Translated real estate client requirements into technical specs, POC prototyping, work breakdown structures, and effort/risk estimates to keep releases on schedule.

ML Engineer | mVizn, Singapore | Remote Full time **JAN 2024 – AUG 2025**

- Created a scalable end-to-end Machine Learning system from training to deployment (**AWS CodeDeploy, EC2**) for 3D semantic segmentation of indoor point clouds extracted from LiDAR scans using **Python** and **PyTorch** reducing Building Information Modeling (BIM) creation time by 60%.
- Designed relational databases using **PostgreSQL** (**Amazon RDS**) enabling logging of user and model inference activities.
- Implemented scalable backend architectures using RESTful API (**FastAPI**), data validation using **Pydantic**, asynchronous system design using **asyncio**, and multi-threaded programming, increasing concurrency by 40%.
- Improved reliable code development and reduced testing time by 40% by planning vigorous testing using **PyTest**.
- Organized code reviews, mentoring sessions on CI/CD, deep learning model development enhancing team productivity.

Research Engineer | Institute of Neurosciences, India | Onsite **JAN 2022 – DEC 2023**

- Planned and implemented custom 3D Computer Vision solutions for neurological disorder analysis (**PyTorch**, Python, **OpenCV**), improving diagnosis speed by 40% and achieving 88.68% accuracy on clinical scans.
- Built a robust data ingestion pipeline with **NumPy** and **Pandas**, and designed analytics workflows using **matplotlib** and **Streamlit** to enhance stakeholder visibility and trust.
- Optimized deep learning models with **ONNX** and deployed via **Docker** on **AWS (SageMaker, EC2, EKS, IAM)**, enabling scalable training and reducing inference latency by 65%.

Machine Learning Engineer | Mercedes-Benz Research & Development India, India | Onsite **AUG 2018 – AUG 2019**

- Developed and optimized a computer vision pipeline for Vulnerable Road User (VRU) detection by fine-tuning **YOLO**v3 on large-scale annotated datasets, attaining detection mAP50 of 83% at real-time inference speeds; designed & deployed model training, evaluation, and validation workflows on **AWS**.
- Designed a normalized **MySQL** database schema and built RESTful APIs with **FastAPI**, tested via **Postman** and integrated with an internal API Gateway, for an interactive dashboard tracking data annotation and QA progress — improving concurrency by 73% using **asyncio** and multi-threading.

Assistant Systems Engineer | Tata Consultancy Services, India | Onsite **AUG 2014 – APR 2015**

- Maintained high-availability backend systems (Java, MySQL) for TCS's internal website, serving 100,000+ users; streamlined the feature release cycle, reducing deployment time by 20%.

EDUCATION

PhD | Jadavpur University, India	FEB 2020 – APR 2025
M.Tech. Computer Science & Engineering | Jadavpur University, India | GPA - 8.83	AUG 2016 – JUN 2018
B.Tech. Computer Science & Engineering | Techno India, India | GPA - 8.81	AUG 2010 - MAY 2014

SKILLS

Languages:	Python, Java, SQL
LLM/Agent Systems:	Multi-Agent Orchestration (LangGraph, CrewAI), LangChain, Retrieval Augmented Generation (RAG), Model Context Protocol (MCP), Prompt Engineering, Local LLMs (Ollama), Agent Evaluation (Ragas, DeepEval), Observability/Tracing (LangSmith), Agentic Memory, LLM APIs (OpenAI, Gemini, Claude)
Frameworks / Libraries:	FastAPI, Flask, Pydantic, asyncio, RESTful API, Postman, PyTest, MySQL, PyTorch, Scikit-learn, OpenCV, Open3D, Pandas, NumPy
Cloud & Ops:	Google Cloud Platform (Cloud Run, Cloud Storage, IAM/RBAC), AWS (SageMaker, EC2), Git, Docker, MLflow, GitHub Actions
Delivery & Leadership:	Code Review, Mentoring, Technical Architecture Ownership, Work Breakdown & Effort Estimation, Risk Assessment, Stakeholder & POC Presentations

PROJECTS

Job Application Agent | https://github.com/arijitde92/Job_Application_Agent

- Developed an autonomous multi-agent workflow using **CrewAI** for career automation, orchestrating specialized agents with carefully chosen different LLM models for repository analysis, profile engineering, and interview readiness.
- Engineered a semantic retrieval layer leveraging **Weaviate** vector capabilities and **Vogaye AI** embeddings to perform high-fidelity indexing of codebase intelligence.
- Built a production-grade full-stack platform with **FastAPI**, managing data via **Google Cloud SQL** and **GCS**, while enabling real-time telemetry of agent logic via Server-Sent Events (SSE) to a React frontend.
- Integrated advanced toolsets for lead scraping through Bright Data MCP and automated summarization using LangChain for sophisticated content processing and ETL.

Doctor Appointment Scheduler | https://github.com/arijitde92/Doctor_Appointment_Scheduler

- Built a multi-agent conversational appointment-booking system (**Google ADK**) with a root coordinator delegating to doctor-matching, review, and scheduling agents.
- Integrated MCP-based tool access for structured data and external actions — **Cloud SQL MCP** for doctor/clinic/availability lookups, **Google Maps MCP** for travel-time estimation, and **Google Calendar MCP** (via Composio) for appointment booking.
- Implemented grounded retrieval over doctor review data using **Vertex AI RAG Engine** and Gemini 2.5 Flash for conversational reasoning.
- Deployed end-to-end on Google Cloud Run.

CERTIFICATIONS

- Agentic AI: Build Your First Agentic AI System | LinkedIn | 2026 | Verify here
- Foundation: Introduction to LangGraph - Python | LangChain Academy | 2026 | Verify here
- Foundation: Introduction to LangChain - Python | LangChain Academy | 2026 | Verify here
- Google Cloud Gen AI Academy | Google and Hack2Skill | 2026 | Verify here
- Multi AI Agent Systems with crewAI | CrewAI | 2025 | Verify here
- Deep Learning, a 5-course specialization | Deeplearning.ai | Verify here
- MCPS: Microsoft Certified Professional, Microsoft Certification No.: 1042C4-5DHF30